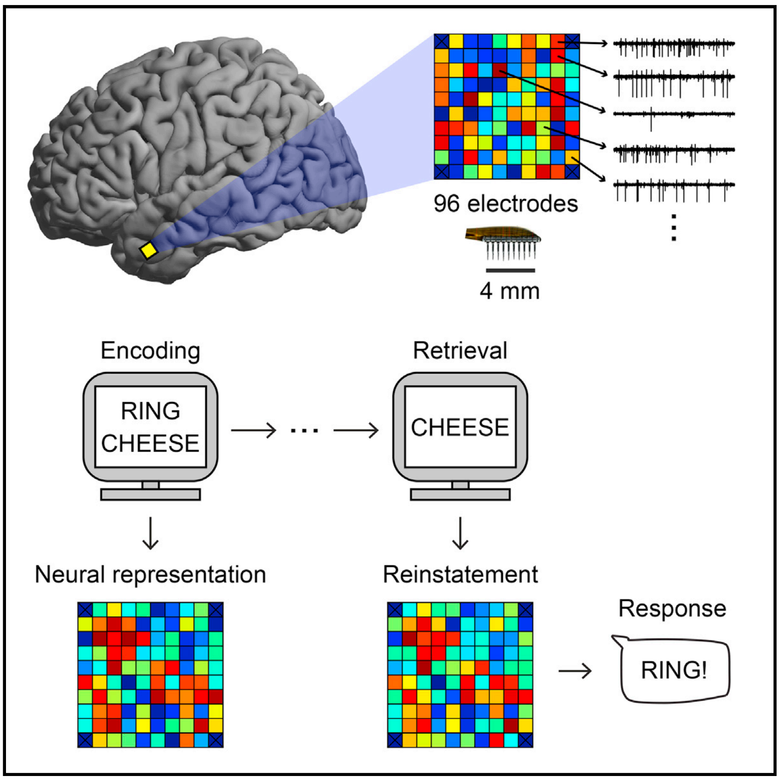
Current Biology

Report

Human Cortical Neurons in the Anterior Temporal Lobe Reinstate Spiking Activity during Verbal Memory Retrieval

Graphical Abstract



Highlights

- Memory retrieval reinstates spiking activity in the human middle temporal gyrus (MTG)
- MTG neurons show spiking responses when forming associations between pairs of words
- Reinstatement of spiking activity is specific when retrieving verbal associations

Authors

Anthony I. Jang, John H. Wittig, Jr., Sara K. Inati, Kareem A. Zaghloul

Correspondence

kareem.zaghloul@nih.gov

In Brief

Jang et al. use microelectrode arrays to record from neuronal populations in the human middle temporal gyrus and show that unique spiking activity patterns representing words are reinstated when retrieving those words from memory.

Jang et al., 2017, Current Biology *27*, 1700–1705
June 5, 2017 Published by Elsevier Ltd.
http://dx.doi.org/10.1016/j.cub.2017.05.014

CellPress

Current Biology
Report

Human Cortical Neurons in the Anterior Temporal Lobe Reinstate Spiking Activity during Verbal Memory Retrieval

Anthony I. Jang,[1,3] John H. Wittig, Jr.,[1,3] Sara K. Inati,[2] and Kareem A. Zaghloul[1,4,*]
[1]Surgical Neurology Branch, NINDS, NIH, Bethesda, MD 20892, USA
[2]Office of the Clinical Director, NINDS, NIH, Bethesda, MD 20892, USA
[3]These authors contributed equally
[4]Lead Contact
*Correspondence: kareem.zaghloul@nih.gov
http://dx.doi.org/10.1016/j.cub.2017.05.014

SUMMARY

When we recall an experience, we rely upon the associations that we formed during the experience, such as those among objects, time, and place [1]. These associations are better remembered when they are familiar and draw upon generalized knowledge, suggesting that we use semantic memory in the service of episodic memory [2, 3]. Moreover, converging evidence suggests that episodic memory retrieval involves the reinstatement of neural activity that was present when we first experienced the event. Therefore, we hypothesized that retrieving associations should also reinstate the neural activity responsible for semantic processing. Indeed, previous studies have suggested that verbal memory retrieval leads to the reinstatement of activity across regions of the brain that include the distributed semantic processing network [4–6], but it is unknown whether and how individual neurons in the human cortex participate in the reinstatement of semantic representations. Recent advances using high-density microelectrode arrays (MEAs) have allowed clinicians to record from populations of neurons in the human cortex [7, 8]. Here we used MEAs to record neuronal spiking activity in the human middle temporal gyrus (MTG), a cortical region supporting the semantic representation of words [9–11], as participants performed a verbal paired-associates task. We provide novel evidence that population spiking activity in the MTG forms distinct representations of semantic concepts and that these representations are reinstated during the retrieval of those words.

RESULTS

Six participants (four males; age 34.8 ± 11.5 years) with drug-resistant epilepsy participated in a verbal paired-associates task (Figure 1A). Participants studied 400 ± 63 (mean ± SEM) word pairs and successfully recalled 23% ± 5% of words. In every experimental session, participants vocalized more correct words than would be expected by chance ($p < 0.05$, permutation procedure; see the STAR Methods for details). First, we tested whether participants were using the meaning of the words to remember the associations. Across participants, correctly recalled word pairs were more semantically related to each other than words pairs that were not successfully remembered ($t[4] = 3.66$, $p = 0.022$; Figure 1B; see the STAR Methods for details). In addition, when participants responded with an intrusion (incorrect word; 18% ± 6% of trials; intra-list intrusions 2.5% ± 0.007% of trials; Table S1), the cue word was more semantically related to the intruded word than the expected word (Figure 1C). Thus, we see behavioral evidence that participants were using their semantic memory system to facilitate memory encoding and retrieval.

While participants performed the task, we captured single-unit spiking activity from populations of neurons in the middle temporal gyrus (MTG) using specialized 96-channel microelectrode arrays (MEAs) implanted for research purposes (Blackrock Microsystems; 31 ± 9 isolated units per experimental session; 88 ± 31 unique units per participant; Figures 1D–1G and S1). Single-unit responses to word presentation during the encoding period were heterogeneous, with some units showing an increase in firing rate and others showing a decrease (Figures 2A and S2A–S2D). Overall, 127 of 556 (23%) of units showed a significant response to words ($p < 0.05$, t test), consistent with this area's role in semantic processing.

We next determined the percentage of units across all participants exhibiting a significant difference in spiking activity between correct and incorrect trials during every time window (1-s windows, 100-ms steps; $p < 0.05$, two-sample t test; see the STAR Methods for details). This value increased to above chance level ($p < 0.05$, permutation procedure) during both the encoding and retrieval periods (Figure 2B). Comparing the mean spiking activity between correct and incorrect trials across participants, we found an overall decrease in spike rate for correct trials during encoding, but not during retrieval ($p < 0.05$, permutation procedure; Figure 2C). The differences in mean spike rates between correct and incorrect trials were more apparent when examining just the units with overall decreases during the encoding period (Figure S2D). To test whether this decrease in activity was related to sparse coding, in every trial we computed the proportion of active units within the population

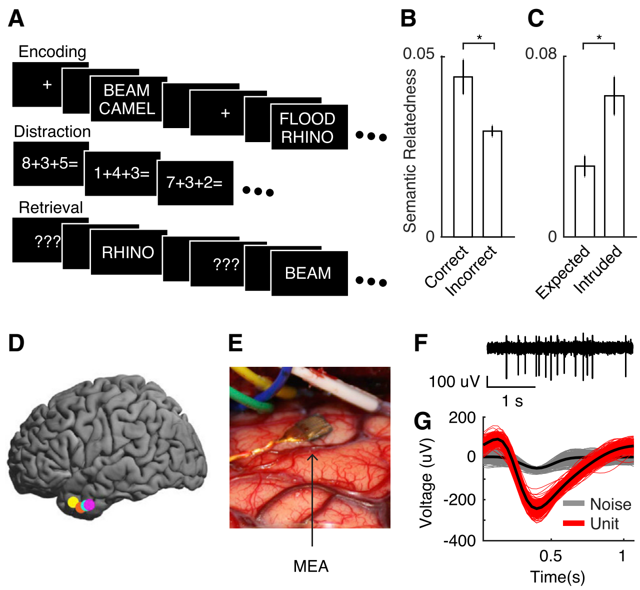


Figure 1. MEA Recordings from the Middle Temporal Gyrus during the Paired-Associates Task

(A) Task presentation. During the encoding period, pairs of words are sequentially presented on the screen. Then, participants engage in an arithmetic distractor task for up to 20 s. During retrieval, one word from each pair is randomly chosen and presented, and participants are instructed to vocalize the associated word.

(B) Semantic relatedness and recall accuracy. Semantic relatedness was quantified using the Word Association Space [12]. Participants were more likely to correctly recall a word pair if the two words were more semantically related to each other.

(C) Semantic relatedness between the intruded word and cue word. When participants made an intrusion, the intruded word was more semantically related to the cue word than the expected word (the correct word).

(D) Locations of the MEAs in five participants. Colors represent each participant. One participant with a right hemisphere implant is not shown here (see Figure S1A for localization on each participant's brain).

(E) Intra-operative photo showing the implanted MEA in the MTG.

(F) High-frequency (0.3–3 kHz) time series of the raw signal from one microelectrode.

(G) Action potential waveforms from a single unit (red) and noise (gray) captured from the same microelectrode that produced (F). The mean waveforms are shown as black lines.

See also Figure S1 and Table S1.

(see the STAR Methods for details) [13]. Across all trials, we found that a smaller proportion of units was active during correct compared to incorrect trials ($t[5] = -3.41$, $p = 0.019$; Figures 2D, S2E, and S2F), suggesting that subsequently remembered words were more sparsely coded.

Despite the heterogeneity across the population of units, on average, each individual unit tended to have similar activity between encoding and retrieval when the word was successfully remembered (Figures 2E and S2B). This suggests that successful retrieval involves reinstatement of population spiking activity. To explicitly test this on a trial-by-trial basis, we constructed representations of the distributed spiking activity across all units for every time point, and we generated a precise temporal map of reinstatement between the encoding and retrieval periods of every trial (Figure 3A).

Across participants, we found significantly greater reinstatement of population spiking activity during correct compared to incorrect trials ($p \leq 0.015$, permutation procedure; Figures 3B and 3C). When a word pair was successfully remembered, the population activity 1–3 s into the encoding period was reinstated during the retrieval period about 1 s before a correct response. We designated this time window as our temporal region of interest (tROI) for further analyses (Figure 3C, black outline). This significant difference in spiking reinstatement was robust: consistent effects were present when we examined the data across sessions rather than participants ($p = 0.0095$, 16 sessions; Figure S3A), when we examined retrieval periods time locked to the onset of the cue word ($p \leq 0.015$; Figure S3B), and when we restricted our analysis to include only experimental sessions in which the participants were tested with the full list length (six word pairs; $p \leq 0.015$, permutation procedure; Figure S3C; see the STAR Methods for details).

We next asked whether the strength of reinstatement was related to the semantic relatedness of the words in each correct trial, as might be predicted by our behavioral analysis. We pooled all correct trials from all participants and divided them into three equally sized bins (terciles) according to the semantic relatedness of the words. Although correct trials exhibited both greater reinstatement and greater semantic relatedness than incorrect trials, within the correct trials we found no significant relation between the strength of reinstatement and semantic relatedness ($p > 0.05$; Figures S3D–S3F).

We sought to test whether the observed reinstatement during correct trials was specific to individual word pairs or reflected general encoding processes, such as visual perception or attention. To do so, we shuffled correct and incorrect trial labels and compared the reinstatement between the shuffled and true data (Figures 3D and 3E). We first shuffled the labels of all retrieval periods to empirically generate a null condition. The average reinstatement in the tROI of the original unshuffled data was significantly greater than that of the shuffled data ($t[5] = 4.41$, $p = 0.0070$). We next shuffled the labels of only the correct retrieval periods. If the observed reinstatement reflects general encoding and retrieval processes, then reinstatement using the shuffled correct retrieval periods should be identical to that observed using the original correct trials. We found that the average level of reinstatement in the tROI for shuffled correct trials was significantly less than that of the unshuffled data ($t[5] = 2.70$, $p = 0.043$), whereas it was significantly greater than that of the null condition ($t[5] = 4.41$, $p = 0.0070$). To test whether reinstatement is affected by slow temporal drifts in neural activity, we restricted shuffling to only swap retrieval periods from adjacent correct trials (average 97 ± 19 s apart). Reinstatement was still significantly less than in the original unshuffled data ($t[5] = 4.14$, $p = 0.0090$), but not different than the average reinstatement after shuffling all correct trials ($t[5] = -0.17$, $p = 0.87$). These comparisons suggest that population spiking activity reinstates representations of specific word pairs rather than a general encoding process.

Finally, given the known interaction between spiking activity and the local field potential (LFP) signal [14, 15], we were interested in whether there was also evidence of reinstatement using the micro-LFP signals recorded from the microelectrodes. In this case, we used the micro-LFP voltage trace to calculate the

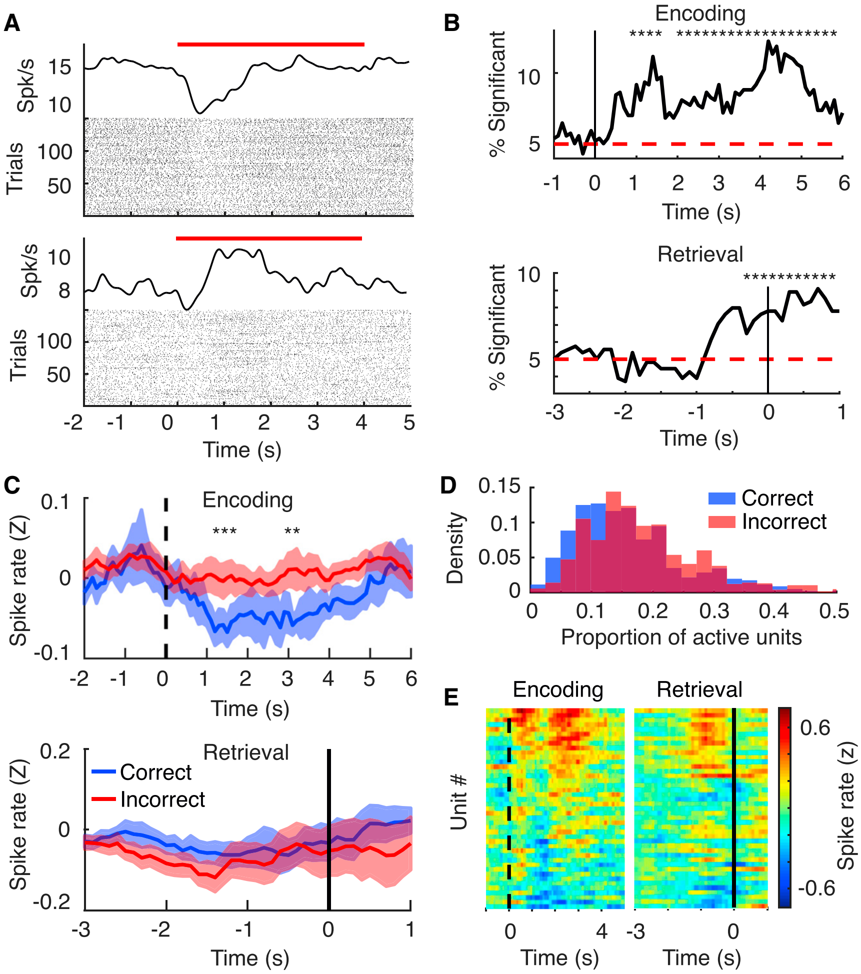


Figure 2. Single-Unit Activity during the Paired-Associates Task

(A) Raster plot and mean spike rate of two example units during the encoding period. Red bars indicate word pair presentation.

(B) Percentage of all units showing a significant difference in spike rates between correct and incorrect trials. Asterisks indicate time points with significant differences ($p < 0.05$, permutation procedure). Red dashed line indicates the 5% line.

(C) Mean Z-scored spike rate of encoding and retrieval periods. Asterisks indicate a significant difference between correct and incorrect trials ($p < 0.05$, permutation procedure; see also Figure S2D). Shaded error bars indicate across-subject SEM.

(D) Distribution of active units during all correct and incorrect encoding periods (see the STAR Methods for details). Correct encoding trials resulted in significantly fewer active neurons than incorrect trials.

(E) Mean Z-transformed spike rate during correct trials for all units recorded during a single experimental session (43 units). Each row represents activity of a single unit. Units tend to show a consistent direction of activity across encoding and retrieval. Rows are ordered based on the mean activity between 0 and 4 s for the encoding period. Dashed lines indicate word pair onset and solid lines indicate response onset.

See also Figure S2.

continuous-time spectral power in each microelectrode in five frequency bands (theta, 3.5–8 Hz; alpha, 8–12 Hz; beta, 13–25 Hz; low gamma, 30–58 Hz; high gamma, 62–100 Hz; see the STAR Methods for details) [6]. Spiking activity during encoding was significantly correlated with high-gamma power ($r = 0.11 \pm 0.016$, $t[5] = 6.95$, $p \leq 0.001$), but not with theta power ($r = 0.011 \pm 0.020$, $t[5] = 0.53$, $p = 0.62$) across participants. Using feature vectors constructed from all five frequency bands, we found significantly greater reinstatement during correct compared to incorrect trials ($p \leq 0.015$, permutation procedure; Figure S3G). Furthermore, we found consistent effects after restricting the feature vectors to only include theta ($p \leq 0.015$, permutation procedure) or high-gamma band activity ($p \leq 0.015$, permutation procedure), two frequency bands that have previously been implicated in successful memory encoding and retrieval [6].

DISCUSSION

Our results provide direct evidence that successfully retrieving associations between words involves reinstatement of population spiking activity in the MTG, a brain region previously highlighted for its role in semantic processing [9–11]. Prior work investigating cortical reinstatement has relied on correlates of neural activity, such as fMRI blood oxygen level-dependent (BOLD) and intracranial electroencephalogram (iEEG) oscillatory activity [4–6, 16]. While these methods provide a useful proxy for neuronal activity, they cannot resolve whether episodic memory retrieval involves the reinstatement of cortical spiking patterns specific to each experience. Our results therefore extend the understanding of the role of the cortex in episodic memory retrieval, by demonstrating reinstatement at the level of individual neurons in the human MTG as participants encode and subsequently retrieve verbal associations.

When encoding an experience into memory, the internally and externally perceived aspects of the experience are initially processed in the cortex and then integrated by the hippocampus into a cohesive memory [17, 18]. Indeed, studies of hippocampal single units in humans have demonstrated that hippocampal activity not only predicts how well word pairs will be remembered [19] but also exhibits stimulus-specific activity [20] and reinstatement during memory retrieval [21, 22]. However, while the hippocampus is engaged when storing and retrieving associative memories [23, 24], ultimately the interaction with cortical sensory and association areas is required to re-experience the event [18, 25]. As such, our data provide evidence for this proposed mechanism of episodic memory retrieval, namely that recall of an experience involves reinstating the relevant cortical neuronal activity that occurred during the initial experience of the event. The reinstatement that we observed here for verbal stimuli may be analogous to reinstatement that has been reported for

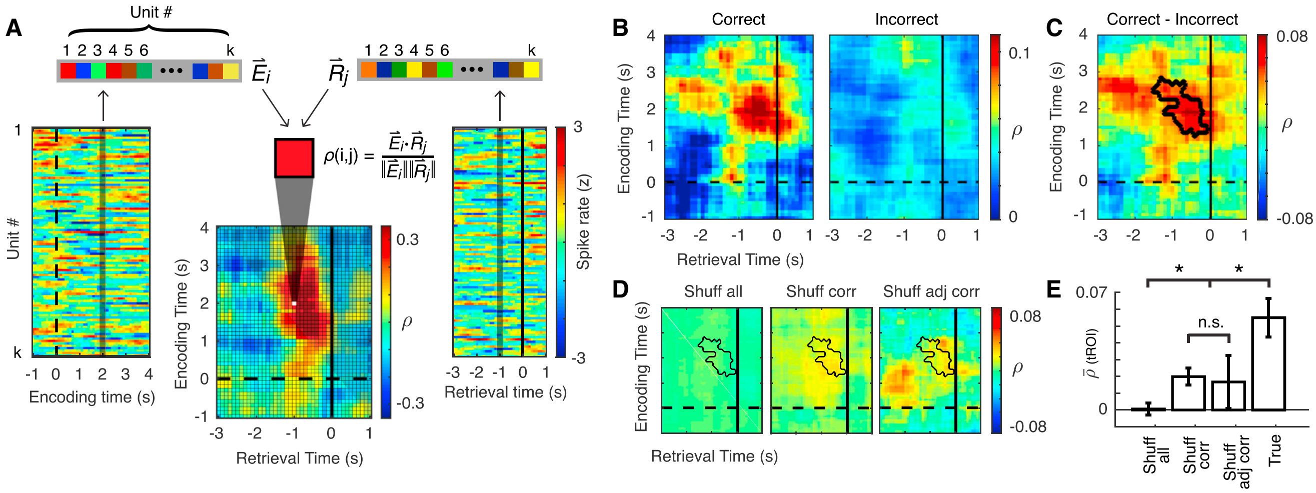


Figure 3. Reinstatement of Neuronal Population Activity

(A) Computing reinstatement for a single trial of encoding and retrieval. The plots on the left and right show population spiking activity for the encoding and retrieval periods, respectively. Each row represents the *Z*-scored spike rate of a single unit across time. For every time window during an encoding and retrieval period, we construct a population vector of the spike rate from every unit (vertical gray bar). The cosine similarity between the encoding (E_i) and retrieval (R_j) population vectors determines the value of a single pixel in the reinstatement map ($\rho[i, j]$).

(B) Reinstatement maps averaged across participants, separately for correct and incorrect trials.

(C) Difference between the mean correct and incorrect reinstatement maps. The region showing a significant difference ($p < 0.05$, permutation procedure) across participants is outlined in black, which was designated as our temporal region of interest (tROI).

(D) Averaged reinstatement maps after each shuffling procedure. Shuff all, difference between correct and incorrect trials after shuffling all retrieval trials; Shuff corr, difference after shuffling all correct retrieval trials; Shuff adj corr, difference after shuffling adjacent correct retrieval trials. For (A)–(D), dashed lines indicate word onset and solid lines indicate response onset.

(E) Mean tROI reinstatement after performing each shuffling procedure. Error bars indicate across-subject SEM.

See also Figure S3.

perceptual stimuli in sensory regions [4, 16]. In this case, however, recalling word pair associations involves reinstatement of semantic representations in the MTG, thus enabling one to retrieve the identity of the missing word.

In the paired-associates task used here, participants form associations between unrelated words, and these associations constitute the individual episodes or experiences that are subsequently recalled. Participants can use many strategies to form these associations. They could associate the low-level perceptual features of the words without drawing upon the meaning of the words (e.g., for DESK and COPPER, they could rehearse a single novel word, DESKCOPPER). Alternatively, encoding these associations may draw upon the meanings of the words in order to form a conceptual link between them using imagery or other semantic strategies that elaborate on these meanings (e.g., visualizing a DESK made of COPPER) [1, 2]. Both our data (Figure 1B) and previous studies [2, 3] support this latter possibility, as participants are more likely to correctly recall word pairs that are more semantically related. Therefore, cortical regions responsible for semantic processing, particularly those that relate written words to their meanings, such as the anterior temporal lobe [9, 10], should be engaged when forming these associations.

Consistent with this, we demonstrate that a significant proportion of units in the human MTG are responsive to the presentation of individual word pairs. Although we observed patterns of neuronal activity specific to individual word pairs, we found that the majority of units exhibit significant decreases in activity during successful encoding, while only some exhibit significant increases. One possible reason may be related to sparse coding, in which an item is represented by the distributed activity of only a small number of neurons [13, 26, 27]. In this case, such a representation could be enhanced by silencing all uninvolved neurons, which would lead to an overall decrease in the population activity. Indeed, fewer neurons were active during correct compared to incorrect encoding periods (Figure 2D), and responses were heterogeneous between neurons for any given word pair and within neurons between individual word pairs. When we separately examined the responses of units exhibiting decreased activity during the encoding period, we found stronger decreases in spiking activity during correct trials compared to incorrect trials (Figure S2D). These data suggest that more effective silencing of overall unit activity during encoding may enhance the sparse spiking representation and, therefore, the memory of the word pair association. Future studies could further elaborate on the existence of sparse coding in this region by examining the population response to repeated presentations of identical stimuli.

Given the variety of strategies that may be deployed when trying to link a pair of words, however, it is possible that the observed cortical activity does not precisely reflect the representation of an individual semantic concept. Instead, such activity could reflect a combination of the two words or any semantic representation the participant utilized internally to facilitate encoding of the word pair. Our data cannot distinguish between these possibilities. Nevertheless, we found reinstatement of neuronal responses unique to each individual word pair, suggesting that a specific semantic representation is engaged and stored for future use in the MTG as new associations are formed. Of note, we did not find a significant relationship between the strength of reinstatement and the semantic relatedness of each correctly recalled word pair. Our data therefore suggest that, while it may be easier to form associations between semantically related words, once formed, each association has a specific neural representation in the brain that is reinstated when it is recalled, regardless of how easy or hard it was to initially form that association.

Our data also suggest that the reinstatement observed here may not be particular only to episodic memory. In fact, successful memory formation in our task may also involve working memory. The list length used here falls within the range of working memory capacity [28, 29], even with the presence of an intervening distractor task [30]. When restricting our analysis to those sessions that involved the full list length and a longer distracting interval, however, we still observed significantly greater reinstatement of spiking activity during correct compared to incorrect trials. Hence, our data suggest that neural reinstatement is a feature of long-term memory retrieval but also raise the possibility that reinstatement of neural activity may be a more general process that is invoked whenever a memory is retrieved.

We previously reported distributed reinstatement of iEEG oscillatory activity across multiple cortical regions, as well as localized reinstatement of iEEG gamma activity in the anterior temporal lobe [6]. Here we show that the same region of the anterior temporal lobe also exhibits reinstatement at the level of individual neurons and micro-LFP, demonstrating that this is a robust phenomenon that spans multiple spatial scales. Distributed reinstatement could reflect communication between brain regions [31], whereas localized gamma activity in iEEG and micro-LFP likely represents population-averaged spike rates in the electrode's vicinity [14]. Here we found heterogeneous responses and sparse coding attributable to the activity of individual neurons, which may not be evident in high-gamma activity alone even at the level of micro-LFP signals. These additional features of neuron-level reinstatement may be important for the representation of individual semantic concepts. Of note, examining both spiking and iEEG activity in participants with drug-resistant epilepsy raises the possibility that the observed responses here may be affected by the underlying pathological process. Although we removed all trials affected by transient ictal activity and based our findings only on within-participant comparisons between correct and incorrect trials, our data need to be interpreted in the context of other studies of memory in the healthy brain.

In summary, our findings provide new insights into the role of the human anterior MTG in episodic memory encoding and retrieval at the neuronal level. We show that the reinstatement of spiking activity in the MTG is unique to the association formed between each pair of words. Given the role of the MTG in semantic processing, our data therefore suggest that forming these associations involves a specific semantic representation for each word pair, and, more generally, that this region may play a role in recalling the semantic content and overall meaning of our past experiences.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- CONTACT FOR REAGENT AND RESOURCE SHARING
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
- METHOD DETAILS
  - Paired-associates task
  - Identification of single units
  - Spike preprocessing
  - Single-unit spiking characteristics
  - Single-unit spiking activity during the paired-associates task
  - Computing reinstatement of population spiking activity
  - Computing reinstatement of micro-LFP spectral power
- QUANTIFICATION AND STATISTICAL ANALYSIS
  - General
  - Identifying chance performance
- DATA AND SOFTWARE AVAILABILITY

SUPPLEMENTAL INFORMATION

Supplemental Information includes three figures and one table and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2017.05.014.

AUTHOR CONTRIBUTIONS

A.I.J., J.H.W., and K.A.Z. conceptualized the study. A.I.J., J.H.W., S.K.I., and K.A.Z. developed the methodology and collected the data. A.I.J. analyzed the data. A.I.J., J.H.W., S.K.I., and K.A.Z. wrote the manuscript. K.A.Z. acquired funding and provided supervision.

ACKNOWLEDGMENTS

We would like to thank Vishnu Sreekumar, Arne Ekstrom, and Tora Bonnevie for valuable comments about the manuscript. We would also like to thank John Cocjin for assistance in recording single-unit spiking activity, Rob Franklin and the technical support from Blackrock Microsystems, Abhik Ray-Chaudry for assistance with histological sections, Timothy Sheehan for assistance in collecting data, and Michael Trotta for assistance in localizing the MEAs. This work was supported by the Intramural Research Program of the National Institute for Neurological Disorders and Stroke. We are indebted to all patients who selflessly volunteered their time to participate in this study.

Received: February 9, 2017
Revised: April 4, 2017
Accepted: May 4, 2017
Published: May 25, 2017

REFERENCES

1. Kahana, M.J., Howard, M.W., and Polyn, S.M. (2008). Associative retrieval processes in episodic memory. In Learning and Memory: A Comprehensive Reference, J.H. Byrne, ed. (Oxford: Elsevier), pp. 467–490.
2. Madan, C., Glaholt, M., and Caplan, J. (2010). The influence of item properties on association-memory. J. Mem. Lang. *63*, 46–63.
3. Caplan, J.B., and Madan, C.R. (2016). Word imageability enhances association-memory by increasing hippocampal engagement. J. Cogn. Neurosci. *28*, 1522–1538.
4. Johnson, J.D., and Rugg, M.D. (2007). Recollection and the reinstatement of encoding-related cortical activity. Cereb. Cortex *17*, 2507–2515.

5. Manning, J.R., Polyn, S.M., Baltuch, G.H., Litt, B., and Kahana, M.J. (2011). Oscillatory patterns in temporal lobe reveal context reinstatement during memory search. Proc. Natl. Acad. Sci. USA *108*, 12893–12897.
6. Yaffe, R.B., Kerr, M.S.D., Damera, S., Sarma, S.V., Inati, S.K., and Zaghloul, K.A. (2014). Reinstatement of distributed cortical oscillations occurs with precise spatiotemporal dynamics during successful memory retrieval. Proc. Natl. Acad. Sci. USA *111*, 18727–18732.
7. Schevon, C.A., Weiss, S.A., McKhann, G., Jr., Goodman, R.R., Yuste, R., Emerson, R.G., and Trevelyan, A.J. (2012). Evidence of an inhibitory restraint of seizure activity in humans. Nat. Commun. *3*, 1060.
8. Chan, A.M., Dykstra, A.R., Jayaram, V., Leonard, M.K., Travis, K.E., Gygi, B., Baker, J.M., Eskandar, E., Hochberg, L.R., Halgren, E., and Cash, S.S. (2014). Speech-specific tuning of neurons in human superior temporal gyrus. Cereb. Cortex *24*, 2679–2693.
9. Binder, J.R., Desai, R.H., Graves, W.W., and Conant, L.L. (2009). Where is the semantic system? A critical review and meta-analysis of 120 functional neuroimaging studies. Cereb. Cortex *19*, 2767–2796.
10. Ralph, M.A.L., Jefferies, E., Patterson, K., and Rogers, T.T. (2017). The neural and computational bases of semantic cognition. Nat. Rev. Neurosci. *18*, 42–55.
11. Zhao, Y., Song, L., Ding, J., Lin, N., Wang, Q., Du, X., Sun, R., and Han, Z. (2017). Left anterior temporal lobe and bilateral anterior cingulate cortex are semantic hub regions: Evidence from behavior-nodal degree mapping in brain-damaged patients. J. Neurosci. *37*, 141–151.
12. Steyvers, M., Shiffrin, R.M., and Nelson, D.L. (2004). Word association spaces for predicting semantic similarity effects in episodic memory. In Cognitive Psychology and its Applications: Festschrift in Honor of Lyle Bourne, Walter Kintsch, Thomas Landauer, and A.F. Healy, eds. (Washington, DC: American Psychological Association).
13. Willmore, B., and Tolhurst, D.J. (2001). Characterizing the sparseness of neural codes. Network *12*, 255–270.
14. Manning, J.R., Jacobs, J., Fried, I., and Kahana, M.J. (2009). Broadband shifts in local field potential power spectra are correlated with single-neuron spiking in humans. J. Neurosci. *29*, 13613–13620.
15. Ray, S., and Maunsell, J.H.R. (2011). Different origins of gamma rhythm and high-gamma activity in macaque visual cortex. PLoS Biol. *9*, e1000610.
16. Deuker, L., Olligs, J., Fell, J., Kranz, T.A., Mormann, F., Montag, C., Reuter, M., Elger, C.E., and Axmacher, N. (2013). Memory consolidation by replay of stimulus-specific neural activity. J. Neurosci. *33*, 19373–19383.
17. Mayes, A., Montaldi, D., and Migo, E. (2007). Associative memory and the medial temporal lobes. Trends Cogn. Sci. *11*, 126–135.
18. Preston, A.R., and Eichenbaum, H. (2013). Interplay of hippocampus and prefrontal cortex in memory. Curr. Biol. *23*, R764–R773.
19. Cameron, K.A., Yashar, S., Wilson, C.L., and Fried, I. (2001). Human hippocampal neurons predict how well word pairs will be remembered. Neuron *30*, 289–298.
20. Gelbard-Sagiv, H., Mukamel, R., Harel, M., Malach, R., and Fried, I. (2008). Internally generated reactivation of single neurons in human hippocampus during free recall. Science *322*, 96–101.
21. Miller, J.F., Neufang, M., Solway, A., Brandt, A., Trippel, M., Mader, I., Hefft, S., Merkow, M., Polyn, S.M., Jacobs, J., et al. (2013). Neural activity in human hippocampal formation reveals the spatial context of retrieved memories. Science *342*, 1111–1114.
22. Howard, M.W., Viskontas, I.V., Shankar, K.H., and Fried, I. (2012). Ensembles of human MTL neurons "jump back in time" in response to a repeated stimulus. Hippocampus *22*, 1833–1847.
23. Wirth, S., Yanike, M., Frank, L.M., Smith, A.C., Brown, E.N., and Suzuki, W.A. (2003). Single neurons in the monkey hippocampus and learning of new associations. Science *300*, 1578–1581.
24. De Falco, E., Ison, M.J., Fried, I., and Quian Quiroga, R. (2016). Long-term coding of personal and universal associations underlying the memory web in the human brain. Nat. Commun. *7*, 13408.
25. Sederberg, P.B., Schulze-Bonhage, A., Madsen, J.R., Bromfield, E.B., Litt, B., Brandt, A., and Kahana, M.J. (2007). Gamma oscillations distinguish true from false memories. Psychol. Sci. *18*, 927–932.
26. Olshausen, B.A., and Field, D.J. (2004). Sparse coding of sensory inputs. Curr. Opin. Neurobiol. *14*, 481–487.
27. Vinje, W.E., and Gallant, J.L. (2000). Sparse coding and decorrelation in primary visual cortex during natural vision. Science *287*, 1273–1276.
28. Conway, A.R., Kane, M.J., and Engle, R.W. (2003). Working memory capacity and its relation to general intelligence. Trends Cogn. Sci. *7*, 547–552.
29. Copeland, D.E., and Radvansky, G.A. (2001). Phonological similarity in working memory. Mem. Cognit. *29*, 774–776.
30. Yoon, J.H., Curtis, C.E., and D'Esposito, M. (2006). Differential effects of distraction during working memory on delay-period activity in the prefrontal cortex and the visual association cortex. Neuroimage *29*, 1117–1126.
31. Buzsáki, G. (2006). Rhythms of the Brain (New York: Oxford University Press).
32. Saad, Z.S., and Reynolds, R.C. (2012). Suma. Neuroimage *62*, 768–773.
33. Fraser, G.W., and Schwartz, A.B. (2012). Recording from the same neurons chronically in motor cortex. J. Neurophysiol. *107*, 1970–1978.
34. Joshua, M., Elias, S., Levine, O., and Bergman, H. (2007). Quantifying the isolation quality of extracellularly recorded action potentials. J. Neurosci. Methods *163*, 267–282.
35. Vaz, A.P., Yaffe, R.B., Wittig, J.H., Jr., Inati, S.K., and Zaghloul, K.A. (2017). Dual origins of measured phase-amplitude coupling reveal distinct neural mechanisms underlying episodic memory in the human cortex. Neuroimage *148*, 148–159.
36. Barthó, P., Hirase, H., Monconduit, L., Zugaro, M., Harris, K.D., and Buzsáki, G. (2004). Characterization of neocortical principal cells and interneurons by network interactions and extracellular features. J. Neurophysiol. *92*, 600–608.
37. Kihlberg, J.K., Herson, J.H., and Schotz, W.E. (1972). Square root transformation revisited. J. R. Stat. Soc. Ser. C Appl. Stat. *21*, 76–81.
38. Maris, E., and Oostenveld, R. (2007). Nonparametric statistical testing of EEG- and MEG-data. J. Neurosci. Methods *164*, 177–190.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms		
MATLAB	Mathworks	https://www.mathworks.com/
Adobe Illustrator	Adobe	https://www.adobe.com/Illustrator
Offline Sorter	Plexon	http://www.plexon.com/products/offline-sorter
Other		
Cerebus System (microelectrode acquisition system)	Blackrock Microsystems	http://blackrockmicro.com/
Cereplex I (Utah Array)	Blackrock Microsystems	http://blackrockmicro.com/
EEG-1100 (video-EEG acquisition system)	Nihon Kohden	http://www.nihonkohden.com/

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources may be directed to, and will be fulfilled by, the Lead Contact Kareem Zaghloul (kareem.zaghloul@nih.gov).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Six participants (4 male; age 34.8 ± 11.5 years) with medication-resistant epilepsy underwent a surgical procedure in which platinum recording contacts (PMT Corporation, Chanhassen, MN, USA) were implanted subdurally on the cortical surface as well as deep within the brain parenchyma. For research purposes, we placed one or two 96-channel microelectrode arrays (MEA; 4 × 4 mm, Cereplex I; Blackrock Microsystems, Salt Lake City, UT, USA) in the middle temporal gyrus (MTG) of each participant in addition to the subdural grid (Figures 1D, 1E, and S1). MEAs were implanted only in participants with a presurgical evaluation indicating clear seizure localization in the temporal lobe. Hence, the implant site in the MTG was chosen to fall within the expected resection area. Each MEA was placed in an area of cortex that appeared normal both on the pre-operative MRI and on visual inspection. Across participants, MEAs were implanted 14.6 ± 3.7 mm away from the closest iEEG electrode with any ictal or interictal activity identified by the clinical team. Four out of the six participants received or are planning to receive a surgical resection which includes the tissue where the MEAs were implanted. One participant had evidence of focal cortical seizure activity and received a localized resection posterior to the MEA site. One participant did not have a sufficient number of seizures during the monitoring period to justify a subsequent resection.

Microelectrodes were arranged in a 10 × 10 grid with each electrode spaced 400 μm apart and extending 1.5 mm into the cortical surface (1.0 mm for one participant). Post-operative paraffin blocks of the resected tissue demonstrated that the electrodes extended approximately halfway into the 3 mm-thick gray matter (Figure S1B). We identified the location of each MEA on a surface reconstruction created using each participant's pre-operative T1 weighted MRI (FreeSurfer, http://surfer.nmr.mgh.harvard.edu). Individual participant reconstructions were co-registered with a standard template brain [32], and the locations of each participant's MEA were visualized on the template brain (Figure 1D). The NINDS Institutional Review Board (IRB) approved the research protocol, and we obtained informed consent from patients explicitly for the placement of the MEAs.

METHOD DETAILS

Paired-associates task

In the paired-associates verbal memory task, participants were sequentially shown a list of word pairs (encoding period) and then later cued with one word from each pair selected at random (retrieval period), during which they were instructed to say the associated word into a microphone (Figure 1A). A single experimental session consisted of 25 lists, where each list contained six pairs of common nouns shown on the center of a laptop screen. Each word pair was preceded by an orientation stimulus ('+') that appeared on the screen for 250 ms, followed by a blank interstimulus interval (ISI) between 500-750 ms. Word pairs were presented on the screen for 4000 ms followed by a blank ISI of 1000 ms. Following the presentation of the list of word pairs, participants completed an arithmetic distractor task of the form A + B + C = ? for 20 s. During the retrieval period, the timing of orientation stimulus ('?????') and word presentation was identical to that of the encoding period. We manually designated each recorded response as correct, intrusion, or pass. A response was designated as pass when no vocalization was made or when the participant vocalized the word 'pass'. We defined all intrusion and pass trials as incorrect trials.

A single session contained 150 total word pairs, or trials. Participants vocalized correct responses with a mean response time of 1878 ± 83 ms, and intrusions with a mean response time of 2577 ± 108 ms. For the remaining trials, participants either made no response to the cue word or vocalized the word 'pass' with a mean response time of 2637 ± 631 ms. For trials with no response, we assigned a response time that was randomly sampled from the distribution of all response times for that session. For three participants with low performance on the standard paired-associates task, we also collected sessions where the number of word pairs per list was reduced to either 2 (75 total lists) or 3 (50 lists), and the math distractor was shortened to a single equation, in order to approach a balanced number of correct and incorrect trials (see Table S1). On average, the time interval between the encoding and retrieval period of the same trial was 23.0 ± 2.4 s for the shortened sessions, and 58.4 ± 0.46 s for the full-length sessions. Each participant completed between 1-4 sessions (20 total sessions across all participants). We excluded four sessions due to low trial counts (< 10 correct or incorrect trials) or low unit counts (< 5 units). We defined each trial epoch as 1 s before to 4 s after the word onset during encoding, and 3 s before to 1 s after vocalization during retrieval.

We computed the semantic relatedness between each pair of words using the Word Association Space (WAS) [12]. WAS provides a database in which words are represented by 400-element vectors, computed from free association norms involving more than 5,000 words and 6,000 participants. We did not compute semantic relatedness for one participant who completed a Spanish version of the task (therefore, the degrees of freedom are 4 instead of 5 for comparisons using WAS). We quantified the semantic relatedness between two words in a pair by computing the cosine similarity between the two word vectors in this multidimensional semantic space.

Identification of single units

We digitally recorded microelectrode signals at 30 kHz using a Cerebus acquisition system (Blackrock Microsystems), with 16-bit precision and a range of ± 8 mV. To extract neuronal spiking activity, we re-referenced each electrode's signal offline by subtracting the mean signal of all the electrodes in the MEA, and then used a second order Butterworth filter to band pass the signal between 0.3 to 3kHz. Using a spike-sorting software package (Plexon Offline Sorter, Dallas, TX, USA), we identified spike waveforms by manually setting a negative or positive voltage threshold depending on the direction of putative action potentials. The voltage threshold was set to include noise signals used in calculating unit isolation quality (see below). Waveforms (duration, 1.067 ms; 32 samples per waveform) that crossed the voltage threshold were stored for spike sorting. Spike clusters were manually identified by viewing the first two principal components (PC1 versus PC2; Figure S1C), and the difference in peak-to-trough voltage (voltage versus time; Figure S1D) of the waveforms. We manually drew a boundary around clusters of waveforms that were differentiable from noise throughout the experimental session. In this manner, we initially identified a total of 1469 putative single units.

Chronic extracellular electrodes have the potential to record from the same unit for multiple days. Therefore, one cannot assume that units recorded in different experimental sessions represent independent samples. To address this, we implemented a metric of unit identification to determine whether any units recorded across two consecutive experimental sessions were the same [33]. The metric assumes that distinct units can be separated using four characteristics of spiking activity: 1) waveform shape, 2) autocorrelation of the spike times, 3) mean firing rate, and 4) pairwise cross-correlation of spike times between units in the same recording session. Each pair of units recorded across two consecutive sessions was therefore represented by a single point in four-dimensional space using the four metrics above. Each four-dimensional similarity score, computed using a pair of units from different experimental sessions, can represent the similarity of either two units on different electrodes, which cannot be identical, or two units recorded from the same electrode, which may or may not be identical. We used a quadratic classifier to determine the decision boundary between different-electrode and same-electrode clusters in this four-dimensional space. This method identified 62 units that were putatively recorded for multiple sessions. The low number of overlapping units is expected, as sessions were on average at least two days apart (76 ± 19 hr), and because over such long periods, units may appear or disappear from recordings due to micromotion, changes in brain state, or other factors [33]. We removed these 62 units from all subsequent analyses to ensure independent samples across sessions.

Due to variability in the signal quality across recordings and the subjective nature of spike sorting, we quantified the quality of each unit by calculating an isolation score and signal to noise ratio (SNR) [34]. The isolation score quantifies the distance between the spike and noise clusters in a 32-dimensional space, where each dimension corresponds to a sample in the spike waveform. The spike cluster consisted of all waveforms that were classified as belonging to that unit, and the noise cluster consisted of all waveforms that crossed the threshold that were not classified as belonging to any unit (Figures S1C and S1D). The isolation score is normalized to be between 0 and 1, and serves as a measure to compare the isolation quality of all units across all experimental sessions and participants. 1200 units (82%) had an isolation score of at least 0.9 (Figure S1E).

In addition to isolation quality, we computed the SNR for each unit using the following equation:

$$SNR = \frac{V_{peak} - V_{trough}}{Noise * C} \tag{1}$$

where V_{peak} and V_{trough} are the maximum and minimum voltage values of the mean waveform, and C is 23 a scaling factor (set as 5) [34]. To obtain *Noise*, we subtracted the mean waveform from each individual waveform for each identified unit, concatenated these waveform residuals, and then computed the standard deviation of this long vector. Therefore, the noise term quantifies the within-unit variability in waveform shape. 1207 units (82%) had an SNR of at least 1.0. We additionally characterized the mean spike rate for each unit, and found that 914 units (62%) had a mean spike rate of at least 1 Hz. Overall, we retained a total of 556 units (38%) that had an isolation score of at least 0.9, a SNR of at least 1.0, a mean spike rate of at least 1 Hz, and were not duplicates across sessions (Figures S1E–S1G). The average firing rate of all units was 3.24 ± 0.11, and the firing rate for the selected units was 3.75 ± 0.55 Hz.

Spike preprocessing

During an experimental session, system-level noise or transient epileptiform activity affecting all electrodes may be present in a small subset of the trials. To identify system-level noise, we computed the mean spike rate across all units during every trial. We sorted the mean spike rates across all trials, and divided the distribution of mean spike rates into quartiles. We identified trial outliers by setting a threshold, $Q3 + w * (Q3 - Q1)$, where $Q1$ and $Q3$ are the mean spike rate boundaries of the first and third quartiles, respectively. We empirically determined w to be 2.8, which most closely matched our criteria for rejecting trials through visual inspection. We excluded all trials with mean spike rates that exceeded this threshold. To identify trials in which transient epileptiform activity may be present, we examined iEEG recordings captured from electrodes overlaid on top of the MEA. In previously published work, epileptiform artifacts were identified by looking for trials displaying excessive variance or kurtosis in the iEEG signal [35]. We calculated and sorted the mean iEEG voltage across all trials, and divided the distribution into quartiles. As with the spiking data, we identified trial outliers by setting a threshold, $Q3 + w * (Q3 - Q1)$, where $Q1$ and $Q3$ are the mean voltage boundaries of the first and third quartiles, respectively. A team of epileptologists empirically determined the weight w to be 2.3. We excluded all trials with mean voltage that exceeded this threshold. The average number of trials removed across all sessions in each participant due to either system-level noise or transient epileptiform activity is shown in Table S1.

Single-unit spiking characteristics

We characterized spiking patterns in this population of units to identify different subgroups of cortical neurons. We computed two waveform metrics commonly used to distinguish different neuronal types - half-amplitude duration and trough to peak time - using the unfiltered waveforms of all retained independent units [36]. Plotting each unit using the above metrics, we did not observe any clear clusters of units (Figure S1H). However, a subset of units did possess the range of values of half-amplitude duration and trough to peak time that were consistent with inhibitory interneurons identified in the rodent neocortex [36], suggesting the existence of interneurons in a small fraction of our dataset.

To identify neurons with bursting activity, we examined the auto-correlograms of spike times for each unit. Bursting activity was identified by large peaks at 3-6 ms followed by exponential decay, while non-bursting activity was identified by exponential rise from time 0. We quantified the 'burstiness' of a unit by dividing the maximum peak of the auto-correlogram between 3-6 ms by the mean value between 7-50 ms. Most units displayed an intermediate burstiness score between the two extreme patterns, and we were not able to identify a clear distinction between the two types of activity (Figure S1I).

Single-unit spiking activity during the paired-associates task

To generate a measure of continuous-time firing rate, we smoothed the spike train for each unit using a sliding window of 1 s with 100 ms steps, and summed the number of spikes for each window. We then square-root transformed the spike counts to stabilize spike noise variance [37]. In order to compare spike rates across all units in all participants, we converted each unit's spike count to a Z-scored spike rate using the mean and standard deviation during the prestimulus window (0-2 s before word onset) across all encoding and retrieval periods.

We determined which units showed an overall response to the task by performing a paired t test comparing prestimulus activity (mean Z-scored rate, 0-2 s before word onset) to activity during word presentation (0-4 s after word onset), and identified 127 units (23%) that showed a significant ($p < 0.05$, t test) change in overall spiking activity. Next, we divided the units into those that show spike rate increases and those that show decreases during the encoding period using the sign of the above t-statistic. 333 units (61%) showed an overall decrease, while 216 (39%) showed an overall increase.

To examine whether word pairs during encoding were sparsely coded by the neuronal population, we computed a metric of sparseness for each encoding period using the population activity recorded during that trial [13]. This metric identifies all units that are active during each trial. During each encoding period (0-4 s relative to word onset), we computed the mean Z-scored spike rate for each unit, and the standard deviation of these means across the population of units. We identified every unit with a spike rate that exceeded the standard deviation of the population in every trial as active. Therefore, on each trial we quantified the proportion of units that responded to that particular stimulus [13]. To compare sparseness between correct and incorrect trials, we pooled all encoding periods and computed the mean sparseness for correct versus incorrect trials within each participant, and performed a paired t test across participants. We excluded trials in which all or none of the units were active for visualization, but the statistical results were upheld both with and without these trials.

Computing reinstatement of population spiking activity

For every time window in each trial, we constructed a population vector comprised of the instantaneous Z-scored firing rate for every unit. For each encoding time window (i) and retrieval time window (j), we defined population vectors as:

$$\overrightarrow{E_i} = [z_1(i) \ldots z_k(i)]$$

$$\overrightarrow{R_j} = [z_1(j) \ldots z_k(j)]$$

where $z_u(i)$ is the Z-transformed firing rate of unit $u = 1...k$ in time window i. We thus created a population vector at each time point that contains the same number of elements as there are unique units.

To quantify reinstatement of neuronal population activity during a trial, we calculated the cosine similarity between the encoding and retrieval population vectors $\overrightarrow{E_i}$ and $\overrightarrow{R_j}$ for all pairs of encoding and retrieval times during that trial (Figure 3A). Thus, for each trial, n, we generated a temporal map of reinstatement values:

$$\rho_n(i,j) = \frac{\overrightarrow{E}_i \cdot \overrightarrow{R}_j}{\overrightarrow{E}_i \overrightarrow{R}_j} \tag{2}$$

where $\rho_n(i,\ j)$ corresponds to the reinstatement of spiking activity across all units between encoding time window i and retrieval time window j during trial n. Each temporal map of reinstatement thus demonstrates the temporal regions in encoding and retrieval during which neuronal population activity was most similar. For every participant, we computed temporal maps of reinstatement separately for all correct and incorrect trials.

Computing reinstatement of micro-LFP spectral power

To obtain micro-local field potential (LFP) signals, we re-referenced each microelectrode's raw voltage data by subtracting the average signal across all microelectrode channels, low-pass filtering the signal at 500 Hz, and resampling at 1,000 Hz. We obtained spectral power at each frequency (2 to 100 Hz, logarithmically spaced) by convolving each continuous time voltage signal with complex valued Morlet wavelets (wavelet number 6) to obtain magnitude and phase information. We then squared and log-transformed the magnitude of the resulting transformed signal to generate a continuous measure of instantaneous power for each frequency.

For every 500 ms time window spaced every 100 ms (80% overlap), we constructed a feature vector comprised of the instantaneous Z-scored power of every channel and every frequency band [6]. For each encoding time window (i) and retrieval time window (j), we defined feature vectors as:

$$\overrightarrow{E_i} = [z_{1,1}(i)\ldots z_{1,F}(i)\ldots z_{L,F}(i)]$$

$$\overrightarrow{R_j} = [z_{1,1}(j)\ldots z_{1,F}(j)\ldots z_{L,F}(j)]$$

where $z_{l,f}(i)$ is the Z-transformed power of electrode $l = 1...L$ at frequency band $f = 1...F$ in time window i. For L electrodes and F frequency bands, we thus create a feature vector at each time window that contains $K = L * F$ features. We computed reinstatement of micro-LFP power using these feature vectors and the same cosine similarity metric used to calculate the reinstatement of population spiking activity. We also computed reinstatement of micro-LFP power using the averaged power across all frequency bands, and found increased reinstatement for correct compared to incorrect trials, although this difference was not statistically significant ($p = 0.30$, permutation procedure).

QUANTIFICATION AND STATISTICAL ANALYSIS

General

To obtain the proportion of units displaying significant differences in spike rates between correct and incorrect trials, for each time window, we performed a two-sample t test (two-tailed) on the instantaneous Z-scored firing rate of each unit between correct and incorrect trials. For each time window, we thus determined the percentage of units that had a p value of less than 0.05. This proportion represents the percentage of units exhibiting a significant difference between correct and incorrect trials in the true dataset. To determine whether this percentage was greater than what would be expected by chance, we used a permutation procedure. We permuted the trial labels 1000 times, and for each permutation we again performed a t test between correct and incorrect trials for every unit and computed the percentage of units that exhibited a p value of less than 0.05. In this manner, we created an empiric distribution for the proportion of units that exhibit a difference between correct and incorrect trials by chance at every time point. For every time window, we compared the true proportion to this empiric distribution, and only designated a time window as statistically significant if the true proportion of significant units was less than 2.5% or greater than 97.5% of the permuted distribution. This comparison also generates a p value for every time window, representing the probability that the proportion of units exhibiting differences between correct and incorrect trials is different than chance. To correct for multiple comparisons across time, we used the nonparametric cluster-based procedure described in detail below.

To assess differences in reinstatement between correct and incorrect trials across participants, we first averaged separate temporal maps of reinstatement for correct and incorrect trials within each participant. Since there were more incorrect trials in each session, we matched the trial counts by reducing the number of incorrect trials via bootstrapping, which should make subsequent t tests less sensitive heteroscedasticity in the data. For each session, we randomly sampled incorrect trials with replacement to match the number of correct trials, then computed the average reinstatement. This procedure was repeated 3000 times, and the mean of these iterations was computed.

We then used a nonparametric cluster-based procedure to assess whether there was a significant difference between the averaged reinstatement maps of the correct and incorrect conditions across participants [38]. This procedure identifies contiguous time windows where reinstatement differs between correct and incorrect trials while controlling the familywise error rate. In addition, this procedure identifies significant differences in reinstatement that are consistent across participants, ensuring that any resulting differences do not arise due to the activity of only one or two participants. Briefly, at each point in the reinstatement map, we computed a *t*-statistic and a p value by using a paired t test on the distribution of correct and incorrect average values across participants. We then randomly permuted the trial labels for the participant-specific averages within each participant independently, resulting in a permutation distribution of 2^n (n = number of participants) possible values that are all equally probable under the null hypothesis. Using this method, each distribution of correct and incorrect average values across participants exists twice in the permutations: once with a positive effect, and once with a negative effect when each participant's trial labels are swapped.

For the true data and for each permutation, we identified contiguous clusters of time points with a p value below 0.05 and the same sign of *t*-statistic (positive or negative). Then, for each cluster, we computed a cluster statistic as the sum of the *t*-statistics across all points in that cluster. In this manner, large magnitude cluster statistics can arise from large differences in reinstatement between correct and incorrect trials that extend over a short duration, or from moderate differences that persist over a longer duration. We saved the largest magnitude cluster statistic from each permutation, thus generating an empiric distribution of maximum cluster statistics that would arise by chance. Importantly, we exclude the largest true cluster from this empiric distribution, although the empiric distribution does include the mirror opposite of the largest true cluster where all participant trial labels were switched. We then calculated the two-tailed p value for each cluster observed in the true dataset by counting the total number of empiric clusters with a magnitude greater than the observed magnitude. With 6 participants, the lowest possible resolution of p value is 1/64, or 0.015. Clusters were determined to be significant and corrected for multiple comparisons if their p value calculated in this manner was less than 0.05.

We used a similar nonparametric cluster-based procedure to assess the statistical difference between the averaged spike rate time series for correct compared to incorrect trials. In this case, we identified significant clusters along the single dimension of time to correct for the multiple comparisons in time. We also used the same cluster-based procedure to correct for multiple time point comparisons when assessing the proportion of units exhibiting significant differences in spike rates between correct and incorrect trials. In this case, we calculated a cluster statistic by identifying every time point that exhibited a significant p value using the permutation procedure described above. We converted each p value to a *Z* score by taking the inverse of the cumulative normal distribution using the mean and standard deviation of the permuted distribution. We therefore generated a cluster statistic for each contiguous region of time exhibiting significant effects by summing the *Z* scores from each of the included time points. We compared the true cluster statistic to the distribution of largest magnitude clusters statistics from each permutation to determine whether the true cluster was significantly different than this empiric distribution.

Identifying chance performance

Although the paired-associates task is designed to examine how well each participant remembers individual associations between pairs of words, it is possible that during retrieval a participant could recall and vocalize any word they have previously seen rather than the specific association corresponding to the cued word. In this case, all of the words presented in a given list are available for recollection, and the participant could vocalize a correct word with the appropriate cued word simply by freely recalling that word by chance. To identify the boundary for such chance performance, we performed a simulation for each experimental session in which we determined the number of correct trials that would occur if each participant freely recalled any of the previously seen words in the session. In each session, we identified all trials in which the participant vocalized a word that was previously seen in the session regardless of accuracy, including those from prior lists. This represents all correct, intra-list, and extra-list intrusion trials. For each of these identified trials, we defined the probability of freely recalling and saying the correct word at random as 1/*X*, where X is the number of word pairs per list. For instance, if each list consisted of 6 word pairs, this would mean a 1/6 chance of saying the correct word on each of these trials. Note that this is the most conservative test of chance performance since, in principle, each trial's chance performance should take into account all the words that were previously seen both in the current list (12 words) and in the entire experimental session, in which case the chance of randomly choosing the correct word should be far less than 1/X. In a single simulated session, we therefore obtained the total number of correct trials that would arise by chance. We repeated this procedure 1000 times to generate an empiric distribution of chance performance, and compared the true performance with this distribution of chance performance to generate a p value representing the probability that the true performance was indistinguishable from chance. A session exceeded chance performance if the true performance was greater than the 95th percentile of the chance performance distribution ($p < 0.05$). Using this strict criterion, we found that performance in all sessions were greater than chance (Table S1).

DATA AND SOFTWARE AVAILABILITY

Processed data of population spiking activity used in this study can be found at: https://neuroscience.nih.gov/ninds/zaghloul/downloads.html.